

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing
and Big Data
Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. Vishnu Vardhan Reddy(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Introduction

A media company planning to migrate its video streaming platform to the cloud must design an infrastructure capable of serving millions of concurrent users without interruptions or buffering. Since user traffic changes significantly during movie releases, live sports events, or new content launches, the cloud architecture must be highly scalable, fault-tolerant, and cost-efficient. Cloud computing provides elasticity, load balancing, content delivery networks (CDNs), and automatic resource management, making it ideal for large-scale video streaming platforms.

Cloud Architecture Design

The architecture should include the following components:

- Global DNS Service
- Load Balancer

- Auto-Scaling Group
- Application Servers
- Content Delivery Network (CDN)
- Cloud Object Storage
- Database Service
- Monitoring and Logging Services

Users first connect to the **Load Balancer**, which distributes incoming requests across multiple application servers. The video files are stored in **Cloud Object Storage**, while frequently accessed videos are cached in a **Content Delivery Network (CDN)** located close to users. Auto-scaling automatically adds or removes virtual machines depending on traffic demand.

Scalability Techniques

Cloud platforms support **horizontal scaling**, where additional servers are automatically added during periods of heavy demand. During low traffic, unnecessary servers are removed to reduce costs. Auto-scaling policies are generally based on CPU utilization, memory usage, network traffic, or the number of active users.

Caching Techniques

Caching significantly improves streaming performance.

The company can implement:

- CDN Edge Caching
- Browser Caching
- Application Cache
- Database Query Cache

A CDN stores copies of video files on edge servers located worldwide. When users request videos, they are delivered from the nearest edge server, reducing latency and buffering.

Load Balancing

Load balancing distributes incoming requests evenly among multiple application servers.

Benefits include:

- Prevents server overload
- Improves application availability
- Reduces response time
- Automatically redirects traffic if one server fails
- Supports high availability

Different load balancing algorithms include:

- Round Robin
- Least Connections
- Weighted Round Robin
- IP Hash

Auto-Scaling

Auto-scaling continuously monitors resource utilization.

It automatically:

- Adds servers during traffic spikes
- Removes unused servers during low traffic
- Maintains application performance

- Reduces infrastructure costs

For example, during a live cricket final, millions of users may join simultaneously. Auto-scaling launches additional instances within minutes to maintain uninterrupted streaming.

Cost Optimization Strategies

Cloud environments provide several methods to reduce operational costs.

These include:

- Pay-as-you-go pricing
- Reserved Instances
- Spot Instances
- Auto-scaling
- Lifecycle storage policies
- CDN caching to reduce bandwidth costs
- Storage tiering for old videos

Older videos can be moved to cheaper archival storage, while frequently accessed content remains in high-performance storage.

Monitoring Tools

Monitoring services continuously observe:

- CPU utilization
- Memory usage
- Disk performance
- Network bandwidth
- Active users
- Response time
- Streaming quality

Examples include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring

Monitoring tools automatically trigger scaling actions and generate alerts when abnormal behavior is detected.

Real-World Example

Netflix is one of the largest cloud-based video streaming platforms. It stores video content in cloud storage and distributes it globally using CDNs. During peak viewing hours, cloud auto-scaling launches additional servers to handle increased demand while load balancers distribute traffic efficiently. This architecture allows Netflix to serve millions of users with minimal buffering.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Introduction

Fintech applications require high availability, security, scalability, and reliability. Deploying applications across multiple cloud providers helps reduce vendor lock-in and improves disaster recovery. However, managing containers consistently across different cloud environments is challenging. Container orchestration platforms such as Kubernetes simplify deployment, scaling, networking, and lifecycle management.

Container-Based Architecture

The application is divided into multiple microservices.

Each service runs inside its own container:

- User Authentication
- Payment Gateway
- Transaction Processing
- Notification Service
- Customer Management
- Reporting Service

Containers package the application along with all dependencies, ensuring consistent execution across cloud providers.

Container Orchestration Using Kubernetes

Kubernetes manages containers automatically.

Its responsibilities include:

- Container deployment
- Scheduling
- Auto-scaling
- Load balancing
- Rolling updates
- Automatic recovery
- Resource allocation

If a container crashes, Kubernetes automatically creates a replacement without manual intervention.

Multi-Cloud Deployment Strategy

Applications are distributed across providers such as AWS, Microsoft Azure, and Google Cloud.

Benefits include:

- High availability
- Better disaster recovery
- Reduced vendor dependency
- Geographic redundancy
- Business continuity

Traffic can be directed to the nearest healthy cloud region.

Service Discovery

Containers communicate using Kubernetes Service Discovery.

Each service receives a DNS name, allowing applications to locate each other automatically without hardcoding IP addresses.

Benefits include:

- Simplified communication
- Dynamic networking
- Automatic endpoint updates

Scaling

Kubernetes supports:

- Horizontal Pod Autoscaling
- Vertical Scaling
- Cluster Autoscaling

Scaling decisions are based on CPU utilization, memory usage, or custom application metrics.

Benefits of Multi-Cloud

- Increased availability
- Disaster recovery
- Better global performance
- Vendor independence
- Regulatory compliance
- Flexible pricing

Risks

- Complex management
- Higher networking costs
- Security consistency challenges
- Data synchronization issues
- Monitoring complexity

Real-World Example

A fintech company may deploy customer authentication on Azure, payment processing on AWS, and analytics on Google Cloud. Kubernetes manages all containers uniformly across providers while ensuring consistent deployment and automatic scaling

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.

Introduction

An e-learning platform stores valuable information such as student records, assignments, videos, examination data, and learning materials. Accidental deletion, cyberattacks, hardware failures, or natural disasters can result in data loss. A comprehensive cloud-native backup and disaster recovery strategy is essential for ensuring business continuity.

Backup Strategy

The platform should implement:

- Automated daily backups
- Incremental backups
- Full weekly backups
- Cross-region replication
- Object versioning
- Snapshot backups

Backups should be stored in multiple cloud regions.

Replication

Cloud providers replicate data automatically across different availability zones and regions.

Benefits include:

- High durability
- Fault tolerance
- Continuous availability

- Fast recovery

Versioning

Versioning stores multiple versions of every file.

If a file is accidentally deleted or modified, administrators can restore previous versions immediately.

Disaster Recovery

The disaster recovery plan should include:

- Backup verification
- Cross-region failover
- Recovery testing
- Automated restoration
- Continuous monitoring

Recovery Time Objective (RTO)

RTO is the maximum acceptable time required to restore services after a failure.

It is reduced by:

- Standby servers
- Automated recovery
- Preconfigured infrastructure

Recovery Point Objective (RPO)

RPO represents the maximum acceptable amount of data loss.

It is minimized using:

- Continuous replication
- Frequent backups
- Real-time synchronization

Business Continuity Steps

1. Enable automatic backups.
2. Store backups in multiple regions.
3. Enable object versioning.
4. Monitor backup jobs.
5. Test disaster recovery regularly.
6. Automate failover.
7. Document recovery procedures.

Example

If ransomware encrypts the primary storage, administrators can restore clean backup versions from another cloud region within minutes, ensuring students continue accessing learning resources.

Conclusion

Cloud-native backup, replication, versioning, and disaster recovery provide reliable data protection, minimize downtime, reduce data loss, and ensure continuous availability of educational services.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Introduction

Government agencies often manage both sensitive and non-sensitive information. Sensitive applications containing citizen records or national security information must remain on-premises due to legal and compliance requirements, while public services such as websites or document portals can be deployed in the public cloud. A hybrid cloud combines both environments securely.

Hybrid Architecture

The architecture consists of:

- On-premises Data Center
- Public Cloud
- Secure VPN or Dedicated Connection
- Identity Management System
- Cloud Storage
- Monitoring Platform

Sensitive applications remain inside government infrastructure, while public-facing applications are hosted in the cloud.

Secure Communication

Communication should use:

- VPN
- Dedicated Private Links
- TLS/SSL encryption
- Firewalls
- Intrusion Detection Systems

These technologies protect data while traveling between environments.

Identity Management

Identity management should implement:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Federated Identity Management

These mechanisms ensure that only authorized personnel can access sensitive systems.

Encryption

Data should be encrypted:

- During transmission
- While stored
- During backups

Encryption keys should be securely managed using cloud key management services.

Challenges

- Infrastructure complexity
- Security management
- Compliance monitoring
- Network latency
- Data synchronization
- Cost management

Conclusion

A hybrid cloud allows government agencies to maintain strict compliance while benefiting from

cloud scalability, flexibility, and cost efficiency

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Introduction

A SaaS provider must maintain continuous service availability even if an entire cloud region fails. Multi-region deployment provides redundancy by hosting identical applications in multiple geographic locations.

Multi-Region Architecture

Each region contains:

- Load Balancer
- Application Servers
- Database
- Storage
- Monitoring Services

Traffic is routed to the nearest healthy region.

Data Replication

Databases should use:

- Synchronous replication
- Asynchronous replication
- Cross-region replication

Replication ensures data consistency and minimizes data loss.

Failover Mechanism

Automatic failover continuously monitors application health.

If one region becomes unavailable:

- Traffic is redirected.
- Secondary databases become primary.
- Applications continue operating without interruption.

Monitoring and Alerting

Monitoring tools track:

- CPU utilization
- Memory usage
- Network performance
- Application health
- Database performance
- Availability

Alerts notify administrators immediately when failures occur.

Benefits

- High availability
- Fault tolerance
- Reduced downtime
- Business continuity
- Better customer experience
- Disaster recovery

Example

A SaaS CRM application deploys identical infrastructure in **Asia, Europe, and North America**. If the Asia region experiences an outage, users are automatically redirected to the nearest available region, ensuring uninterrupted access with minimal service disruption.

Conclusion

A multi-region cloud architecture with data replication, automatic failover, and continuous monitoring ensures maximum availability, fault tolerance, and reliable service delivery. It is the preferred architecture for SaaS providers that require uninterrupted operations and global accessibility.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented